

CNN-Based Crop Disease Diagnosis with Retrieval-Driven LLM Support

Anurag Kumar, Abhinav Yadav, Ayush Katiyar

Supervised by: Mr. Yousuf Haider (Assistant Professor, CSIT)

Computer Science & Information Technology Department
KIET Group of Institutions, Ghaziabad

4 May 2026



Table of Contents

- 1 Introduction
- 2 Problem Statement
- 3 Literature Review
- 4 System Design
- 5 Results and Conclusion
- 6 References

- Crop disease diagnosis remains a challenge due to variable field conditions and limited access to expert agronomists.
- Deep learning models, especially CNNs, achieve high accuracy but are limited to detection tasks. Existing systems lack actionable, context-aware recommendations, creating a gap between prediction and real-world usability.
- To address this, we propose a hybrid CNN–RAG–LLM architecture that integrates visual diagnosis with retrieval-grounded reasoning for end-to-end decision support.

Problem Statement

- Crop diseases are difficult to detect accurately due to changing environmental conditions (humidity, temperature, soil variability).
- In India, crop diseases cause losses of over **50,000–90,000 crore annually**, significantly impacting farmer income and national food security.
- Deep learning models provide high accuracy but are limited to **prediction only**, without advisory support.
- Farmers receive disease labels but lack **actionable, real-time guidance**, creating a gap between AI predictions and real-world usability.

Literature Review

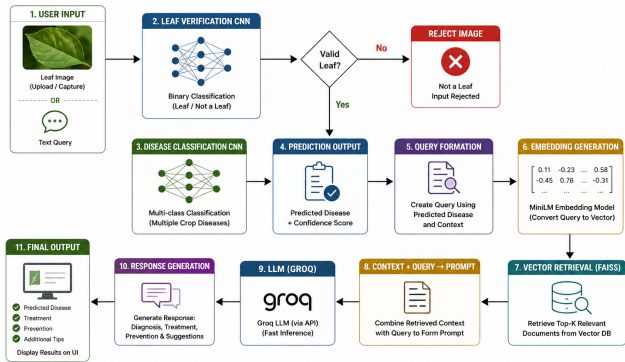
Title	Authors	Year	Journal	Summary
Plant Disease Detection using Deep Learning	Chohan, Murk, et al.	2020	IJRTE	Efficient plant disease detection system leveraging CNNs.
A Review on Machine Learning Classification Techniques for Plant Disease Detection	Shruthi, U., V. Nagaveni, and B. K. Raghavendra	2019	ICDI3C (IEEE)	Comparative study on plant disease classification techniques.
An Overview of the Research on Plant Leaves Disease Detection using Image Processing Techniques	Gavhale, Kiran R., and Ujwala Gawande	2014	IOSR-JCE	Review of plant disease detection using image processing techniques and challenges.
Deep CNN Based Detection System for Real-Time Corn Plant Disease Recognition	Mishra, Sumita, Rishabh Sachan, and Diksha Rajpal	2023	Procedia Computer Science (167)	Deep learning approach for real-time corn disease identification.
Plant Disease Detection Using Machine Learning	Ramesh, Shima, et al.	2024	ICDI3C (IEEE)	Comparative study on plant disease classification methods.

System Design & Data Pipeline

- Dataset consists of labeled crop leaf images covering multiple diseases and crop types for robust multi-class classification.
- Preprocessing includes image resizing, normalization, and data augmentation (rotation, flipping, brightness adjustment) to improve model generalization and reduce overfitting.
- A custom lightweight CNN architecture is designed to efficiently extract relevant features and accurately classify crop diseases.
- The system is optimized for reliable performance under real-world environmental conditions such as varying lighting, backgrounds, and noise.

System Workflow (CropCure AI)

CropCure AI – System Flow Overview



- Image input → leaf verification → disease classification.
- Model outputs predicted disease with confidence score.
- FAISS retrieves relevant agricultural knowledge.
- LLM generates diagnosis, treatment, and prevention.
- Final results displayed to the user.

Results and Conclusion

- Achieved 93.06% accuracy and 92% F1-score, demonstrating strong and reliable classification performance across different crop disease classes.
- CNN–RAG–LLM pipeline provides accurate and context-aware diagnosis by using knowledge-based reasoning.
- Improves reliability and reduces hallucinations using retrieval grounding, ensuring responses are based on verified agricultural information.
- Bridges the gap between prediction and actionable farmer guidance by providing clear, practical recommendations for disease treatment.

- Z. Hossain et al., “Advanced Deep Learning Techniques for Precision Diagnosis of Tea Leaf Diseases,” in *Proc. IEEE Int. Conf. Emerging Technologies and Applications (MPSec ICETA)*, pp. 1–6, 2025.
- A. Iqbal, “PlantHealthNet: Transformer-Enhanced Hybrid Models for Disease Diagnosis and Severity Estimation in Agriculture,” *IEEE Access*, 2025.
- M. E. Rayed, J. R. Jim, M. J. Islam, M. F. Mridha, M. M. Kabir, and M. J. Hossen, “MangoLeafXNet: An Explainable Deep Learning Model for Accurate Mango Leaf Disease Classification,” *IEEE Access*, 2025.
- M. S. A. M. Al-Gaashani, R. Alkanhel, M. A. S. Ali, M. S. A. Muthanna, A. Aziz, and A. Muthanna, “MSCPNet: A Multi-Scale Convolutional Pooling Network for Maize Disease Classification,” *IEEE Access*, 2025.
- S. N. Yousafzai et al., “Multi-Stage Neural Network-Based Ensemble Learning Approach for Wheat Leaf Disease Classification,” *IEEE Access*, 2025.
- A. Bhargava et al., “Plant Leaf Disease Detection, Classification, and Diagnosis Using Computer Vision and Artificial Intelligence: A Review,” *IEEE Access*, vol. 12, pp. 37443–37469, 2024.

References (cont.)

- D. S. Joseph, P. M. Pawar, and K. Chakradeo, "Real-time plant disease dataset development and detection using deep learning," *IEEE Access*, vol. 12, pp. 16310–16333, 2024.
- V. Balafas et al., "Machine learning and deep learning for plant disease classification and detection," *IEEE Access*, vol. 11, pp. 114352–114377, 2023.
- A. Alharbi, M. U. G. Khan, and B. Tayyaba, "Wheat disease classification using continual learning," *IEEE Access*, vol. 11, pp. 90016–90026, 2023.
- L. Goel and J. Nagpal, "A systematic review of recent machine learning techniques for plant disease identification and classification," *IETE Technical Review*, vol. 40, no. 3, pp. 423–439, 2023.
- R. R. Patil and S. Kumar, "Rice-fusion: A multimodality data fusion framework for rice disease diagnosis," *IEEE Access*, vol. 10, pp. 5207–5222, 2022.
- L. Li, S. Zhang, and B. Wang, "Plant disease detection and classification by deep learning—A review," *IEEE Access*, vol. 9, pp. 56683–56698, 2021.
- W.-J. Hu et al., "MDFC-ResNet: An agricultural IoT system to accurately recognize crop diseases," *IEEE Access*, vol. 8, pp. 115287–115298, 2020.
- S. Mishra, R. Sachan, and D. Rajpal, "Deep convolutional neural network based detection system for real-time corn plant disease recognition," *Procedia Computer Science*, vol. 167, pp. 2003–2010, 2020.
- U. Shruthi, V. Nagaveni, and B. K. Raghavendra, "A review on machine learning classification techniques for plant disease detection," in *Proc. ICACCS*, IEEE, 2019.
- S. Ramesh et al., "Plant disease detection using machine learning," in *Proc. ICDI3C*, IEEE, 2018.
- E. Rex, "Plant Disease Dataset," Kaggle, 2018. Available: <https://www.kaggle.com/datasets/emmarex/plantdisease>